

# Back of the Envelope Estimation

Before designing the architecture of a cloud storage system like [Google Drive](#), it is important to estimate the scale of the system. These rough calculations help determine storage requirements, traffic load, and scalability needs.

## Assumptions

We make the following assumptions for the system:

- Total registered users: **50 million**
- Daily Active Users (DAU): **10 million**
- Each user gets **10 GB** of free storage space
- Each active user uploads **2 files per day**
- Average file size: **500 KB**
- Read to write ratio: **1:1**

## Storage Estimation

### Total Storage Allocation

Each registered user receives 10 GB of storage.

So, total allocated storage becomes:

50 million \* 10 GB = 500 Petabytes

Therefore, the system must support approximately **500 PB** of total storage capacity.

This estimate does not include:

- Replication overhead
- Backups
- Metadata storage
- File version history

Actual required storage would be significantly larger in production systems.

## Upload Traffic Estimation

### Upload API Queries Per Second (QPS)

Each active user uploads 2 files daily.

Total uploads per day:

$10 \text{ million} * 2 = 20 \text{ million uploads/day}$

To calculate average upload QPS:

$20,000,000 / (24 * 3600) \approx 240$

So, the average upload API traffic is approximately:

- **240 QPS**

### Peak Upload QPS

Traffic is usually uneven throughout the day.

Assuming peak traffic is 2× the average:

$240 * 2 = 480$

Hence:

- **Peak Upload QPS  $\approx 480$**

## Daily Data Growth

Each active user uploads:

$2 * 500\text{KB} = 1000\text{KB} \approx 1\text{MB/day}$

Total daily uploaded data:

$10 \text{ million} * 1\text{MB} = 10\text{TB/day}$

Thus, the platform grows by approximately:

- **10 TB of new files daily**

This estimate excludes:

- Replication
- Version history
- Cache storage
- Thumbnail generation
- Metadata indexing

## Read Traffic Estimation

Since the read-to-write ratio is **1:1**, download traffic is approximately equal to upload traffic.

Therefore:

- Average download QPS  $\approx$  **240**
- Peak download QPS  $\approx$  **480**

## Key Insights from Estimation

These estimations help us understand several important architectural requirements:

### Massive Storage Requirement

The system must support:

- Petabyte-scale distributed storage
- Data replication
- High durability

## High Scalability

The architecture should:

- Scale horizontally
- Handle increasing file uploads and downloads
- Support millions of concurrent users

## Efficient Synchronization

Since users access files from multiple devices:

- Sync mechanisms must be optimized
- Delta synchronization should be considered

## Bandwidth Optimization

Frequent uploads/downloads require:

- Compression
- Chunked file transfer
- CDN support
- Deduplication techniques